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Verney

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(54) **EXOSKELETON UPPER LIMB WITH
IMPROVED COMPACTNESS**

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(71) Applicant: **COMMISSARIAT A L'ENERGIE
ATOMIQUE ET AUX ENERGIES
ALTERNATIVES**, Paris (FR)

(72) Inventor: **Alexandre Verney**, Grenoble (FR)

(73) Assignee: **COMMISSARIAT A L'ENERGIE
ATOMIQUE ET AUX ENERGIES
ALTERNATIVES**, Paris (FR)

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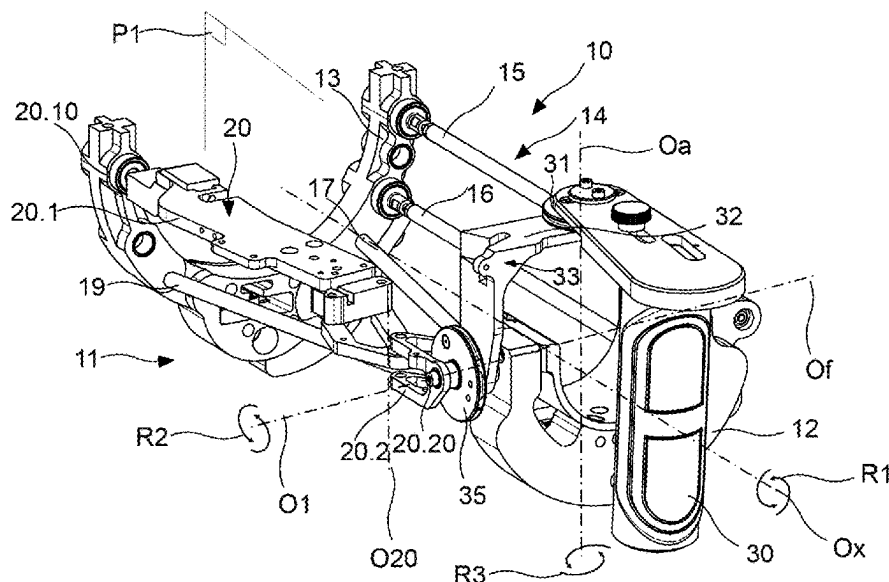
Primary Examiner — Tu A Vo

(74) *Attorney, Agent, or Firm* — Oblon, McClelland,
Maier & Neustadt, L.L.P.

(57) **ABSTRACT**

An exoskeleton upper limb includes a frame and a connection interface movably mounted relative to the frame to perform a second rotation and a third rotation. The second rotation is actuated by a first kinematic chain of actuation including a first actuator. The third rotation is actuated by a second kinematic chain of actuation including a second actuator. The first kinematic chain includes a first connecting rod and/or the second kinematic chain includes a second connecting rod.

10 Claims, 4 Drawing Sheets



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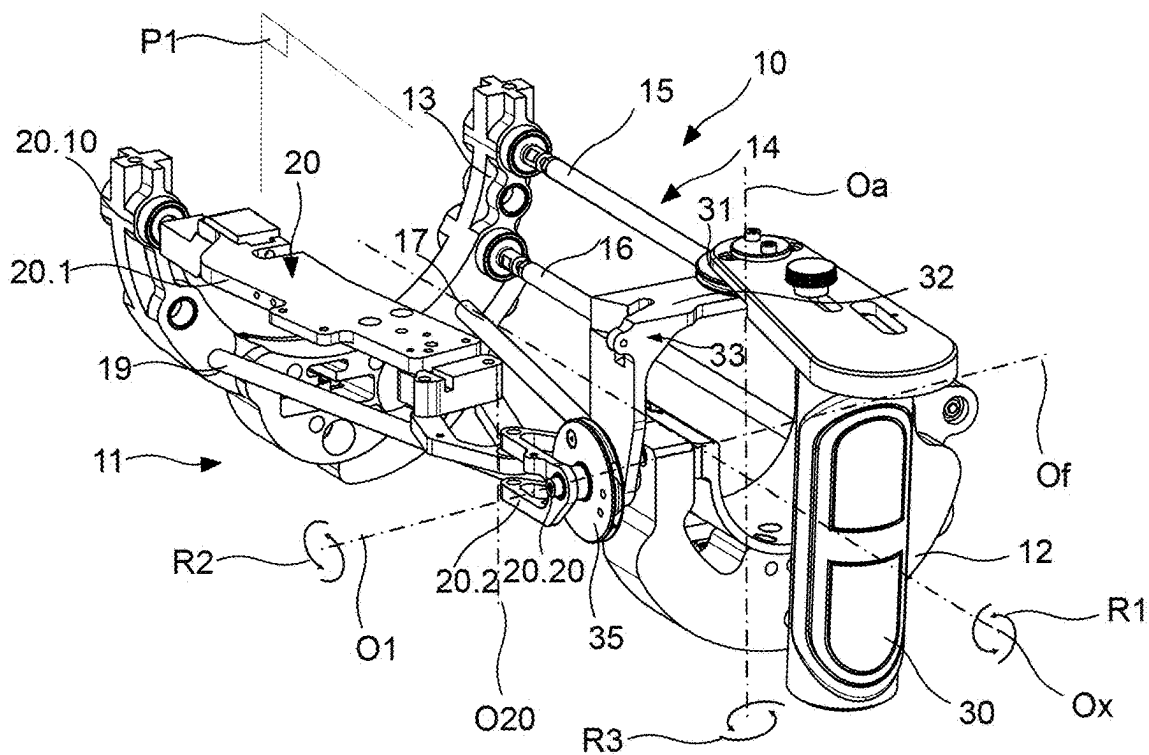
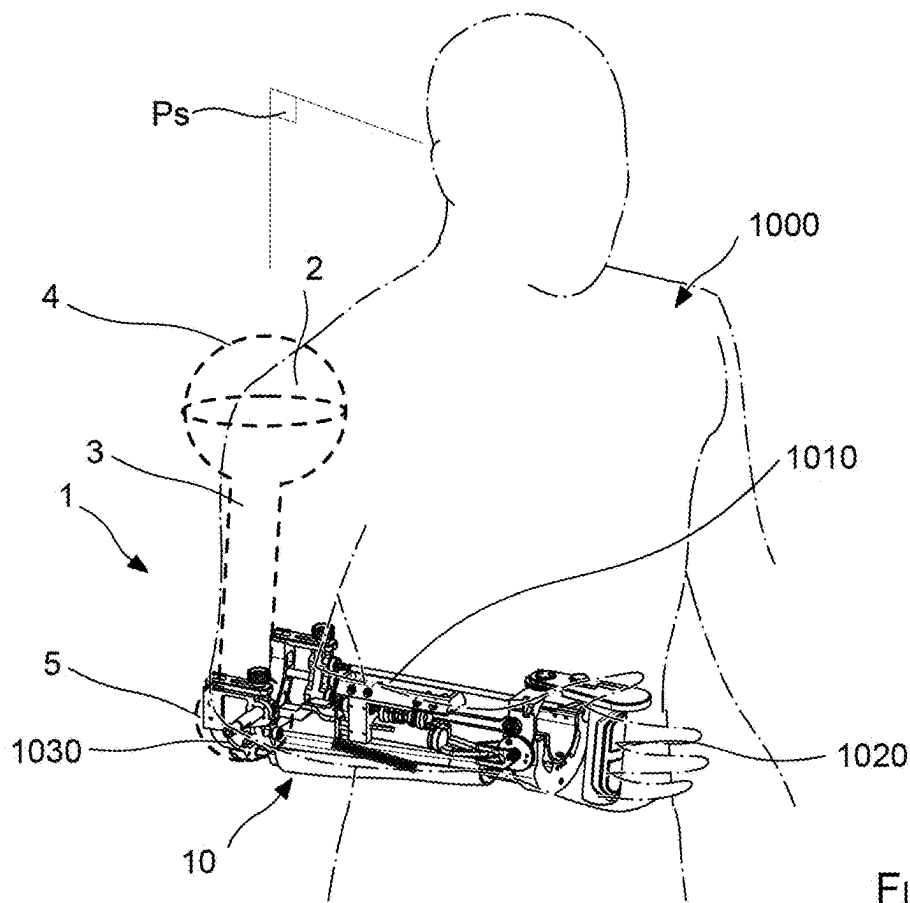
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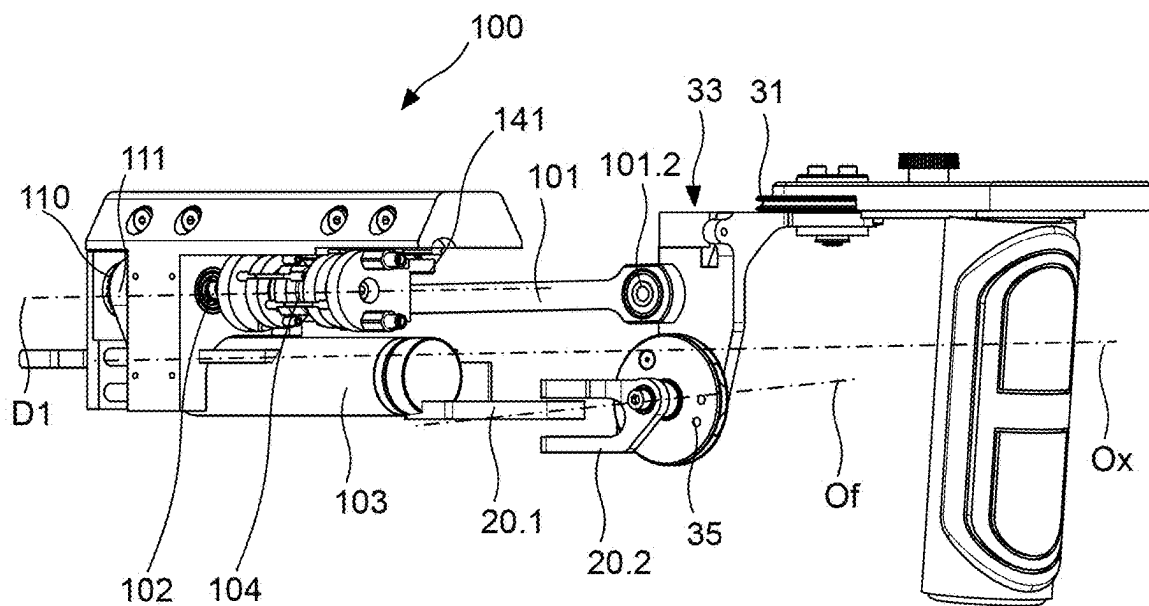
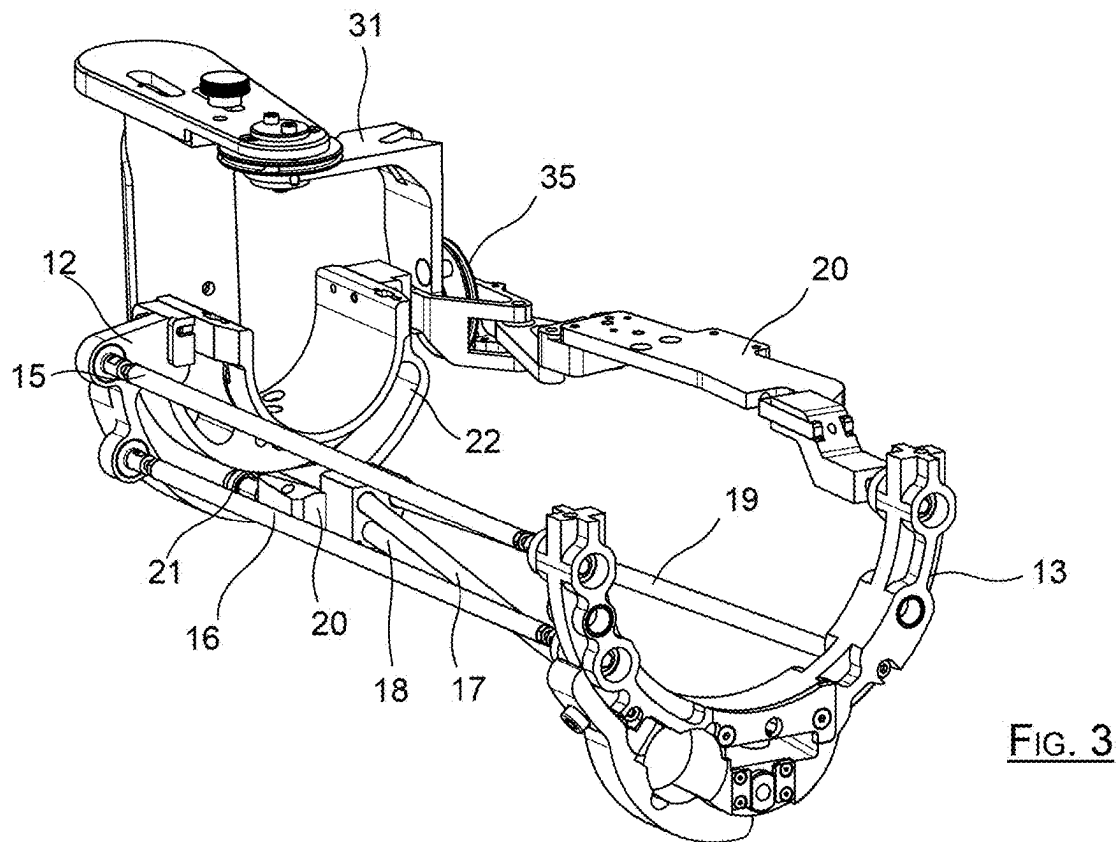
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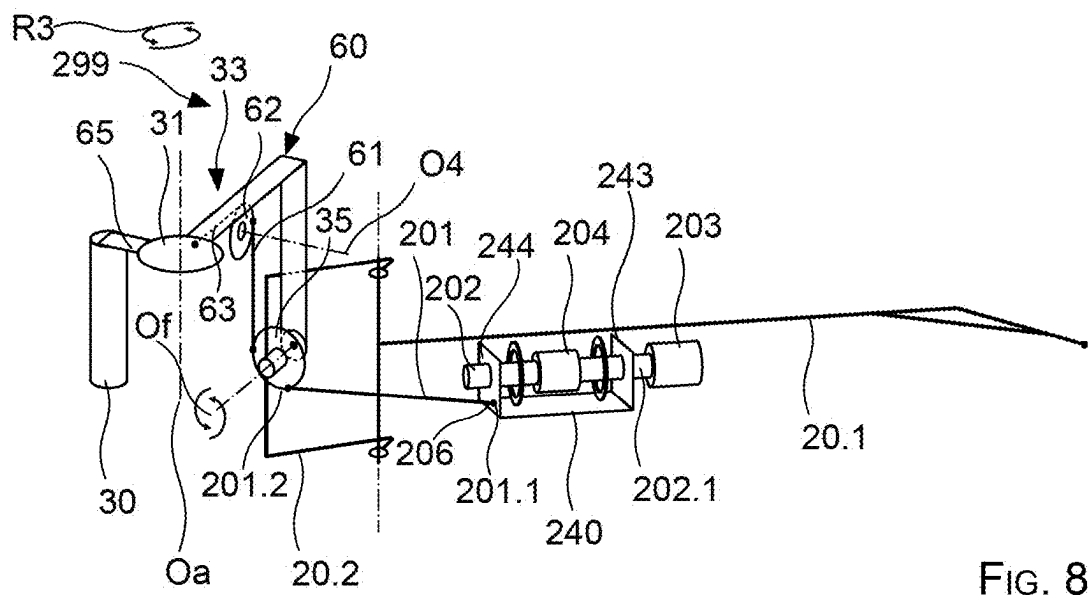
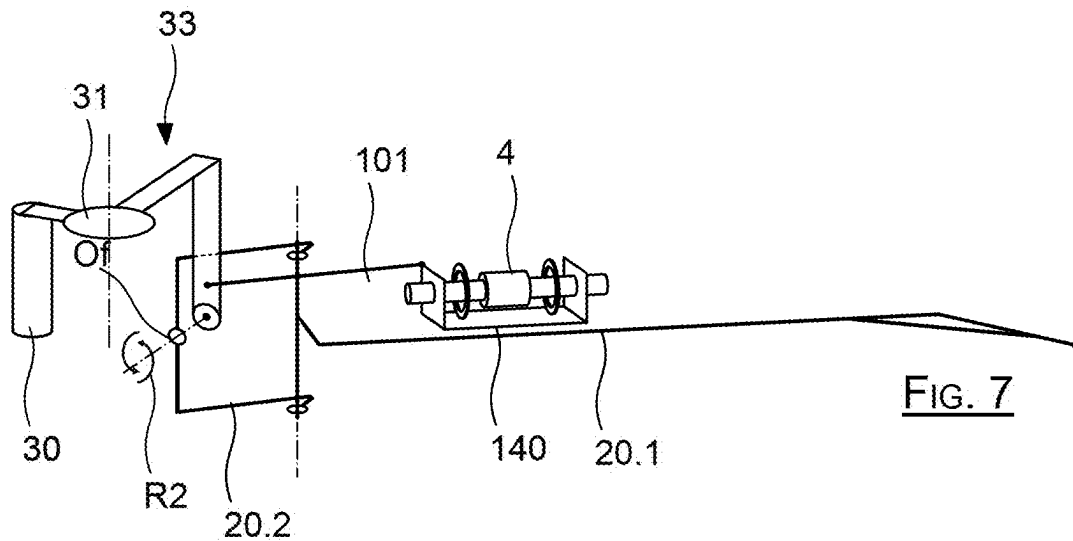
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EXOSKELETON UPPER LIMB WITH IMPROVED COMPACTNESS

TECHNICAL FIELD

The invention relates to the field of exoskeletons and more specifically the field of exoskeleton upper limbs and their actuation.

PRIOR ART

Exoskeletons are devices similar to motorised manipulators and are arranged outside a limb of the human body, generally parallel thereto. They can aid a user by supporting all or part of the weight of their limb (rehabilitation exoskeleton) or by providing an additional force (assistance exoskeleton). They can also be used as a motion sensor for the limb, while eliminating the fatigue required to maintain its balance during various tasks such as tele-operations, clinical studies, etc.

An exoskeleton upper limb conventionally comprises an arm segment at the distal end of which a forearm segment is hinged. The proximal end of the arm segment is generally hinged to a shoulder joint connected to the user.

If the hand movements of the user are to be assisted, the forearm segment can comprise a frame designed to accommodate a forearm of the user. The frame comprises a front wrist support and a rear support connected by a structure to enable a first pronosupination rotation of the front support relative to the rear support. A handle to be grasped by a hand of the user is movably mounted relative to the frame to perform rotation about a flexion/extension axis and rotation about an adduction/abduction axis. These rotations are generally powered by electric geared motors, the rotary output of which is integral in rotation with the handle. One of the two geared motors is thus integral with a support which is rotated by the other of the two geared motors. The result is a bulky, heavy forearm segment with very high inertia.

In prior art mechanisms, the lever arm of the actuators produces a great bending moment on the structure of the frame, which causes very high local stresses on the rotational guidance of the front support, resulting in unwanted friction during movements. Such systems therefore have relatively low payloads, large footprints and complex actuation kinematics, which adversely affect user comfort, possible applications for exoskeleton upper limbs, manufacturing costs and control complexity.

OBJECT OF THE INVENTION

The object of the invention is to reduce the inertia and kinematic complexity of an upper limb comprising wrist actuators.

DISCLOSURE OF THE INVENTION

To this end, an exoskeleton upper limb is provided comprising a frame extending along a longitudinal axis and which is intended to receive a forearm of a user, the frame comprising a front wrist support and a rear support connected by a structure to enable a first pronosupination rotation of the front support relative to the rear support about the longitudinal axis. The upper limb also comprises an interface for connection to a hand of the user movably mounted relative to the frame to perform a second rotation about a flexion/extension axis and to perform a third rotation about an adduction/abduction axis. The second rotation is

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actuated by a first kinematic chain of actuation comprising a first actuator and the third rotation is actuated by a second kinematic chain of actuation comprising a second actuator. According to the invention, the first kinematic chain comprises a first connecting rod, a first rear end of which is hinged to a first output of the first actuator, the first connecting rod comprising a first front end coupled to the connection interface, and/or the second kinematic chain comprises a second connecting rod, a second rear end of which is hinged to a second output of the second actuator and the second connecting rod comprises a second front end coupled to the connection interface.

According to other specific, non-exclusive and optional embodiments of the invention:

the first front end and/or the first rear end and/or the second front end and/or the second rear end comprise a ball joint;

the second front end is connected to a transmission linkage which comprises a third transmission rod for transmission about a third axis of rotation and a fourth transmission rod for transmission about a fourth axis of rotation;

the third axis coincides with the flexion axis;

the connection interface is rotatably mounted about the adduction axis relative to a structure, the structure itself being rotatably mounted relative to the frame about the flexion axis;

the first actuator comprises a first screw/nut assembly comprising a first nut cooperating with a first screw extending in a first direction substantially parallel to the longitudinal axis, a first anti-rotation device for blocking a rotation of the first nut relative to the first screw, a first motor designed to rotate the first screw about the first direction and cause linear movement of the first nut; wherein the first rear end is coupled to the first nut;

the first anti-rotation device comprises a first carriage connected on the one hand to the first nut and on the other hand to the first rear end of the first connecting rod, the first anti-rotation device also comprising a first guiding mechanism for the translation of the first carriage relative to the frame in a direction parallel to the longitudinal axis, and wherein the first carriage is connected to the first nut by a first pair of cables comprising a first cable and a second cable extending on either side of the first screw as well as by a second pair of cables comprising a third cable and a fourth cable extending on either side of the first screw, the first pair of cables and the second pair of cables extending in a direction substantially parallel to the longitudinal axis, the first pair of cables and the second pair of cables being located on either side of a first plane passing through the first nut and which is normal to the first screw;

the first cable and the second cable are coupled to the first nut by a first intermediate support connected to the first nut by a seventh cable and an eighth cable;

the third cable and the fourth cable are coupled to the first nut by a second intermediate support connected to the first nut by a ninth cable and a tenth cable;

the second actuator comprises a second screw/nut assembly comprising a second nut cooperating with a second screw extending in a second direction substantially parallel to the longitudinal axis, a second anti-rotation device for blocking a rotation of the second nut relative to the second screw, a second motor designed to rotate the second screw about the longitudinal axis and cause

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linear movement of the second nut; wherein the second rear end is coupled to the second nut;
the first and/or the second actuator is a linear output actuator;
the first kinematic chain comprises, successively, a linear output, a connecting rod, a crank mounted to rotate relative to the frame and actuated by the connecting rod.

Further features and advantages of the invention will become apparent on reading the following description of a specific non-limiting embodiment of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

Reference will be made to the appended figures in which:
FIG. 1 is a schematic, perspective view of a user wearing an upper limb according to the invention;

FIG. 2 is a partial schematic, perspective view of the upper limb shown in FIG. 1 as seen from front to back;

FIG. 3 is an identical view to that shown in FIG. 2 but as seen from back to front;

FIG. 4 is a partial schematic, perspective view of the first kinematic chain of the upper limb shown in FIG. 1;

FIG. 5 is a schematic view from above of the first kinematic chain shown in FIG. 4;

FIG. 6 is a schematic side view of the first kinematic chain shown in FIG. 5;

FIG. 7 is a kinematic view of the first kinematic chain shown in FIG. 5;

FIG. 8 is a kinematic view of the second kinematic chain of the upper limb shown in FIG. 1.

DETAILED DISCLOSURE OF PARTICULAR EMBODIMENTS

With reference to FIG. 1, the upper limb (generally designated by reference numeral 1) is part of an exoskeleton (not shown) provided with a shoulder joint 2 and an arm segment 3. The proximal end 4 of the arm segment 3 is hinged to the shoulder joint 2. A forearm segment 10 is hinged to the distal end 5 of the arm segment 3.

The forearm segment 10 is intended to receive a forearm 1010 of a user 1000. The longitudinal axis of the forearm 1010 therefore substantially coincides with the longitudinal axis Ox of the segment 10. The terms front and rear will be used by analogy with the front and rear of the forearm 1010. In this way, the front will correspond to the end intended to be on the side of the hand 1020 and the rear will correspond to the end intended to be on the side of the elbow 1030 of the user 1000.

The forearm segment 10 shown in FIG. 2 comprises a frame 11 extending along a longitudinal axis Ox. The frame 11 comprises a front support 12 for the wrist 1040 of the user 1000, a rear support 13 and a structure 14 connecting the front 12 and rear 13 supports. The front support 12 has the shape of a half-ring and is movably mounted relative to the rear support 13.

The structure 14 here is in the form of a lattice which comprises five bars 15 to 19 and an aluminium plate 20. As shown in FIG. 2, the structure comprises three inner bars 15, 16 and 17, a middle bar 18 and an outer bar 19, 16, 17, 18 and 19. The bars 15 to 17 lie between a sagittal plane Ps of the user 1000 and a first plane P1 comprising the longitudinal axis Ox and which is parallel to the sagittal plane Ps when the user 1000 is in a rest position, their forearm 1010 along their body: a high inner bar 15, an intermediate inner bar 16 and a low inner bar 17. The middle bar 18 is located

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in the plane P1. The low outer bar 19 and the plate 20 are located on another side of the plane P1. The rear longitudinal ends of the bars 15 and 16 and of the plate 20 are connected to the support 13 by a ball joint. The rear ends of the bars 17, 18 and 19 are connected to the support 13 by fixed joints—in this case screwed and glued. The bars 17 to 19 thus form a fixed “triangular lattice” type mast which connects the support 12 to the support 13 and absorbs the shear forces applied to the support 12.

The front longitudinal ends of the bars 15 and 16 are also connected to the support 12 by ball joints. The plate 20 is, for its part, connected to the support 13 by a ball joint and to the support 12 by a pivot joint with axis O1 substantially orthogonal to the plane P1.

As shown in FIGS. 2 and 3, the rear support 13 also has the shape of a half-ring, with a larger diameter than that of the front support 12, in order to delimit a frusto-conical inner space approximating the outer contour of the forearm 1010. The forearm segment 10 therefore has substantially the shape of a truncated cone, the bars 15 to 18 of which form generatrices.

The front support 12 is mounted so that it can pivot relative to the rear support 13 about the axis Ox. To do this, the front ends of the bars 17 and 18 are connected to the support 12 via a roller bearing 21 provided with three rollers 22 engaged in a semi-circular groove 23 formed in the support 12. Preferably, one of the rollers 22 is rotatably mounted on an adjustable eccentric shaft to eliminate operating play. This pivoting link can be actuated about the axis Ox in various ways known to a person skilled in the art, such as by motorising the rollers 21.

This link enables a first pronosupination rotation R1 of the front support 12 relative to the rear support 13 about the longitudinal axis Ox.

As shown in FIGS. 2 and 3, a handle 30 is integral with a first pulley 31 mounted so as to rotate about an adduction-abduction axis Oa by means of a bearing (not shown) located at a first end 32 of an L-shaped structure 33. A second end 34 of the structure 33 is mounted so as to rotate relative to the support 12 about a flexion/extension axis Of, which in this case coincides with the axis O1.

The handle 30 can thus perform a second rotation R2 about the flexion/extension axis and a third rotation R3 about the adduction/abduction axis Oa. The second rotation R2 is actuated by a first kinematic chain 100 and the third rotation R3 is actuated by a second kinematic chain 200.

The first kinematic chain 100 comprises a first connecting rod 101 with a first rear end 101.1 and a first front end 101.2.

The second kinematic chain 200 also comprises a second connecting rod 201 with a second rear end 201.1 and a second front end 201.2.

The plate 20 as well as the first kinematic chain 100 and the second kinematic chain 200 will now be described in detail.

The plate 20 comprises a first portion 20.1 and a second portion 20.2 mounted to pivot relative to one another about an axis O20 orthogonal to a plane parallel to the axes Ox and Of. The first portion 20.1 is connected at its rear end 20.10 to the support 13 by a ball joint, and the second portion 20.2 is connected at its end 20.20 by a pivot joint with axis Of to the support 12.

As shown in FIGS. 4 to 6, the first kinematic chain 100 comprises a first screw 102/nut 104 assembly comprising a first nut 104 cooperating with a first screw 102 which extends in a first direction D1 substantially parallel to the longitudinal axis Ox. The first kinematic chain 100 comprises a first anti-rotation device 140 for blocking a rotation

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of the first nut **104** relative to the first screw **102** and forming a first linear output actuator **199**. The anti-rotation device **140** has a carriage **141** mounted for translational movement in a direction parallel to the axis O_x by means of a first guiding mechanism **150**. The mechanism **150** is in this case a prismatic linear guide which comprises a rail **151** rigidly attached to the plate **19** and on which a runner **152** attached to the carriage **141** slides.

The carriage **141** comprises a base **142** which extends substantially parallel to the longitudinal axis O_x and at the ends of which project a first arm **143** and a second arm **144** which thus constitute the linear output of the actuator **199**. The carriage **141** is connected to the nut **104** by a first pair of cables **160** comprising a first cable **161** and a second cable **162** extending on either side of the screw **102** as well as by a second pair of cables **170** comprising a third cable **171** and a fourth cable **172** extending on either side of the screw **102**. As shown in FIG. 6, the first pair of cables **160** and the second pair of cables **170** extend in a direction substantially parallel to the longitudinal axis O_x and the first pair of cables **160** and the second pair of cables **170** are located on either side of a plane passing through the nut **104** and which is normal to the screw **102**. More specifically, a third end **161.1** of the cable **161** and a fourth end **162.1** of the cable **162** are crimped onto the arm **143**. Symmetrically, a fifth end **171.1** of the cable **171** and a sixth end **172.1** of the cable **172** are crimped onto the arm **144**. A seventh end **161.2** of the first cable **161** and an eighth end **162.2** of the second cable **162** are also coupled to the nut **104** by a first intermediate support **190** connected to the nut **104** by a seventh cable **191** and an eighth cable **192**. The first intermediate support **190** is in this case in the shape of a ring and comprises a first hole **190.1** into which the seventh end **161.2** of the cable **161** is crimped. The first intermediate support **190** also comprises a second hole **190.2** diametrically opposite the first hole **190.1** and into which the eighth end **162.2** of the cable **162** is crimped.

The first intermediate support **190** also comprises a third hole **190.3** into which a ninth end **191.1** of the seventh cable **191** is crimped and a sixth hole **190.4** into which the tenth end **192.1** of the cable **192** is crimped. The eleventh end **191.2** of the cable **191** is crimped into a seventh hole **105.1** of a ring **105** attached to the nut **104**, and the twelfth end **192.2** of the cable **192** is crimped into an eighth hole **105.2** of the ring **105**.

In the same way, the thirteenth end **171.2** of the third cable **171** and the fourteenth end **172.2** of the fourth cable **172** are coupled to the nut **104** by a second intermediate support **195** connected to the nut **104** by a ninth cable **196** and a tenth cable **197**. The second intermediate support **195** is in this case in the shape of a ring and comprises a seventh hole **195.1** into which the thirteenth end **171.2** of the cable **171** is crimped. The second intermediate support **195** also comprises an eighth hole **195.2** diametrically opposite the seventh hole **195.1** and into which the fourteenth end **172.2** of the cable **172** is crimped.

The second intermediate support **195** also comprises a ninth hole **195.3** into which the fifteenth end **196.1** of the ninth cable **196** is crimped and a tenth hole **195.4** into which the sixteenth end **197.1** of the tenth cable **197** is crimped.

The seventeenth end **196.2** of the cable **196** is crimped into the seventh hole **5.1** and the eighteenth end **197.2** of the tenth cable **197** is crimped into the eighth hole **105.2**.

A rear end **102.1** of the screw **102** comprises a gear **110** connected by a belt **111** to a rotary output of a first motor **103**.

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As shown in FIGS. 4 to 6, the first kinematic chain **100** comprises a first connecting rod **101**, a first rear end **101.1** of which is hinged to the arm **144** of the carriage **141** by means of a ball joint **106**. The first front end **101.2** of the first connecting rod **101** is connected to the structure **33**, which acts as the crank and on which the handle **30** is mounted by means of a ball joint **107**.

Thus, the first kinematic chain **100** comprises, successively, a linear output **144** of an actuator **199**, a connecting rod **101**, a crank **33** mounted to rotate relative to the frame **11** and actuated by the connecting rod **101**.

In this way, rotation of the output of the motor **103** rotates the screw **102** and causes linear movement of the first nut **104**. This linear movement is transmitted to the carriage **141** and therefore moves the first rear end **101.1** of the connecting rod **101** relative to the plate **20**—and incidentally to the frame **11**—which therefore rotates the handle **30** about the flexion axis O_f via a connecting rod **101**-crank **30** connection. In this way, the motor **103** actuates the second rotation **R2** via a linear output **144** of an actuator **199**.

The kinematic chain **100** and its main elements are kinematically represented in FIG. 7.

The second kinematic chain **200**, represented in FIG. 8 according to the same representation as that in FIG. 7, comprises a second screw **202**/nut **204** assembly comprising a second nut **204** cooperating with a second screw **202** extending in a second direction $D2$ substantially parallel to the longitudinal axis O_x . The second kinematic chain **200** comprises a second anti-rotation device **240** for blocking a rotation of the second nut **204** relative to the second screw **202**, identical to the first anti-rotation device **140**, and forming a second linear output actuator **299**. The second anti-rotation device **240** comprises a carriage **241** provided with two arms **243** and **244** and which is mounted for translational movement in a direction parallel to the axis O_x by means of a second guiding mechanism (not shown). A rear end **202.1** of the screw **202** is connected to a rotary output of a second motor **203**. The second arm **244** thus constitutes the linear output of the actuator **299**.

As shown in FIGS. 8, the second kinematic chain **200** comprises a second connecting rod **201**, a second rear end **201.1** of which is hinged to the arm **244** of the carriage **241** by means of a ball joint **206**. The first front end **201.2** of the second connecting rod **201** is connected to a second pulley **35** hinged about the axis O_f and shown in FIGS. 2 to 6 and 8. The pulley **35** is connected to a transmission linkage **60**. The linkage **60** comprises a third transmission rod **61** for transmission about the axis O_f , which extends between the pulley **35** and a third transmission pulley **62** pivotally mounted on the structure **33** about a fourth axis of rotation $O4$ substantially orthogonal to a plane parallel to the axes O_a and O_f . A fourth connecting rod **63** connects the pulley **62** and the pulley **31** to actuate the third rotation **R3** of the handle **30** about the axis O_a . The handle **30** is connected to the pulley **31** by a connecting rod **65** integral in rotation with the pulley **31**. The assembly formed by the pulley **31** and the connecting rod **65** forms a crank actuated by the connecting rod **63**.

In this way, rotation of the output of the motor **203** rotates the screw **202** and causes linear movement of the nut **204**. This linear movement is transmitted to the carriage **241** and therefore moves the second rear end **201.1** of the connecting rod **201** relative to the plate **19**—and incidentally relative to the frame **11**—which therefore rotates the handle **30** about the abduction axis O_a . In this way, the motor **203** actuates the third rotation **R3**.

Thus, the second kinematic chain **200** comprises a linear output **244** of an actuator **299**, a connecting rod **63**, a crank **65** mounted to rotate relative to the frame **11** and actuated by the connecting rod **62**.

This results in an upper limb, the forearm actuators of which are more compact and less complex than those in the prior art. The number of parts is also reduced which confers a significant advantage in terms of reliability and production costs. Finally, no actuator is subject to the inertia of the other actuator (weight of the motor and motion transformation elements) which also improves the footprint, weight, energy consumption and manufacturing cost of the upper limb **1**.

Of course, the invention is not limited to the embodiments described but encompasses any alternative embodiment falling within the scope of the invention as defined by the claims. In particular,

although in this case the first kinematic chain and the second kinematic chain both comprise a connecting rod, one rear end of which is coupled to an output of the actuator and one front end of which is coupled to the connection interface, the invention is also applicable to only one of the two kinematic chains being fitted with such a connecting rod, the other chain being able to be actuated according to actuating methods known to a person skilled in the art such as geared motors or cable actuators with idler pulleys;

although in this case the bars are connected to the front and rear supports by ball joints, the invention is also applicable to other types of spherical joints, e.g. a joint providing three rotations about three orthogonal axes, such as a universal joint associated with a pivot joint converging at the centre of the universal joint, or a ball joint associated with a pivot joint, or any other equivalent joint or combination of joints;

although in this case the two ends of the bars are connected to the supports by ball joints, the invention is also applicable to other types of connection of the bars to the supports, such as bars having ball joints at the ends connected to the same support, and cardan joints at their other ends. The joints are advantageously uniform on each support. Alternatively, the links between the bars and the supports could be made directly by elastic joints;

although in this case the limb comprises a handle, the invention is also applicable to other types of interface for connection to a hand of a user such as a glove or a strap;

although in this case the first kinematic chain comprises a screw/nut assembly actuated by a motor and connected to an anti-rotation device for forming a linear output actuator, the invention is also applicable to other types of linear output actuators, for example a linear actuator such as a hydraulic, pneumatic or electric actuator, or even a rack and pinion assembly;

although in this case the first rear end of the first connecting rod is hinged to an arm of the carriage, the invention is also applicable to a first rear end hinged to other types of output of the actuator of the second rotation, such as the base of the carriage or to the nut.

The invention claimed is:

1. An exoskeleton upper limb comprising:

a frame extending along a longitudinal axis and the frame is configured to receive a forearm of a user, the frame comprising a front wrist support and a rear support connected by a structure to enable a first pronosupination rotation of the front wrist support relative to the rear support about the longitudinal axis; and

a connection interface configured to connect to a hand of the user, the connection interface being movably mounted relative to the frame to perform a second rotation about a flexion-extension axis and to perform a third rotation about an adduction-abduction axis, the second rotation being actuated by a first kinematic chain of actuation comprising a first actuator, and the third rotation being actuated by a second kinematic chain of actuation comprising a second actuator, wherein

the first kinematic chain comprises a first connecting rod, a first rear end of the first connecting rod is hinged to a first output of the first actuator, the first connecting rod comprising a first front end coupled to the connection interface, and/or

the second kinematic chain comprises a second connecting rod, a second rear end of the second connecting rod is hinged to a second output of the second actuator and the second connecting rod comprises a second front end coupled to the connection interface.

2. The upper limb according to claim **1**, wherein at least one of the first rear end, the first front end, the second rear end, and the second front end comprise a ball joint.

3. The upper limb according to claim **1**, wherein the second front end is connected to a transmission linkage, the transmission linkage comprises a third transmission rod for transmission about a third axis of rotation and a fourth transmission rod for transmission about a fourth axis of rotation.

4. The upper limb according to claim **3**, wherein the third axis coincides with the flexion-extension axis.

5. The upper limb according to claim **1**, wherein the connection interface is rotatably mounted about the adduction-abduction axis relative to a second structure, the second structure being rotatably mounted relative to the frame about the flexion-extension axis.

6. The upper limb according to claim **1**, wherein the first actuator comprises

a first screw and nut assembly comprising a first nut cooperating with a first screw extending in a first direction substantially parallel to the longitudinal axis; a first anti-rotation device to block a rotation of the first nut relative to the first screw; and

a first motor configured to rotate the first screw about the first direction and cause linear movement of the first nut, wherein

the first rear end is coupled to the first nut.

7. The upper limb according to claim **6**, wherein the first anti-rotation device comprises a first carriage connected to the first nut and to the first rear end of the first connecting rod, the first anti-rotation device also comprising a first guiding mechanism for a translation of the first carriage relative to the frame in a direction parallel to the longitudinal axis, and wherein the first carriage is connected to the first nut by a first pair of cables comprising a first cable and a second cable extending on either side of the first screw as well as by a second pair of cables comprising a third cable and a fourth cable extending on either side of the first screw, the first pair of cables and the second pair of cables extending in a direction substantially parallel to the longitudinal axis, the first pair of cables and the second pair of cables being located on either side of a plane passing through the first nut and the plane is normal to the first screw.

8. The upper limb according to claim **7**, wherein the first cable and the second cable are coupled to the first nut by a first intermediate support connected to the first nut by a seventh cable and an eighth cable.

9. The upper limb according to claim 7, wherein the third cable and the fourth cable are coupled to the first nut by a second intermediate support connected to the first nut by a ninth cable and a tenth cable.

10. The upper limb according to claim 6, wherein the second actuator comprises

a second screw and nut assembly comprising a second nut cooperating with a second screw extending in a second direction substantially parallel to the longitudinal axis; a second anti-rotation device to block a rotation of the second nut relative to the second screw; and a second motor configured to rotate the second screw about the longitudinal axis and cause linear movement of the second nut, wherein

the second rear end is coupled to the second nut.

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